AIML Summer Internship 2026, IIHMF, MNNIT Prayagraj

# HOMEWORTH AI

Project Title : **House Price Prediction Using Machine Learning**


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&

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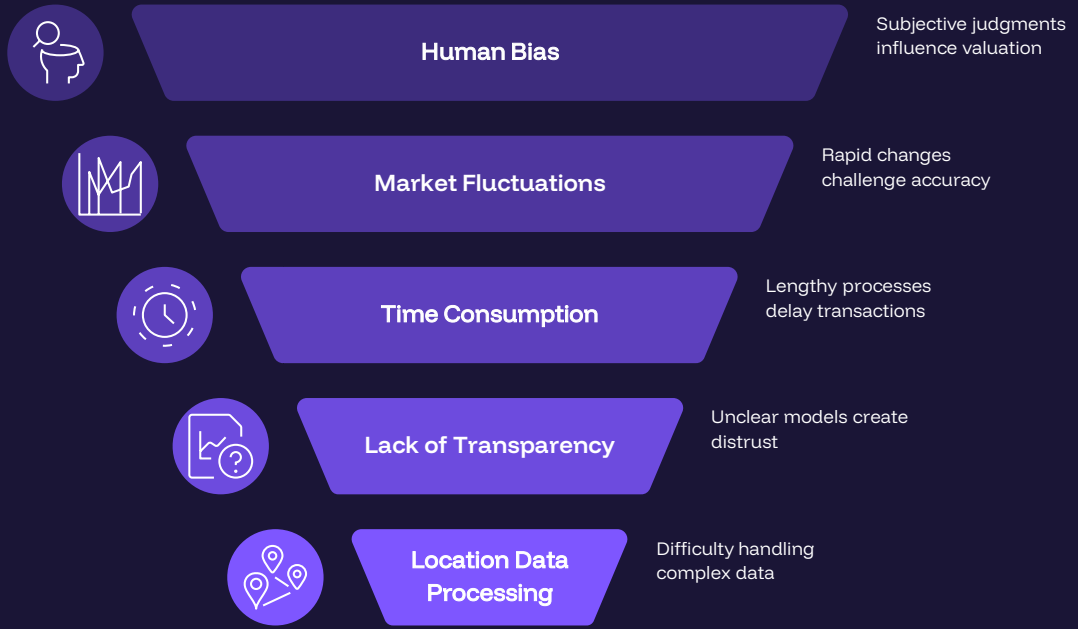
(Project Coordinator)

IIHMF, MNNIT

# The Valuation Gap

## Current challenges in traditional property valuation:

- Manual estimation leads to significant human bias
- Inability to keep pace with rapid market fluctuations
- Time-consuming processes for buyers and sellers
- Lack of transparency in traditional valuation models
- Difficulty in processing multi-dimensional location data



# Project Objectives

01

## Predictive Modeling

Develop high-accuracy regression models to estimate property value based on multi-factor inputs.

02

## Feature Engineering

Transform raw data into meaningful insights like Amenities Score and Prime Location indicators.

03

## Interactive Dashboard

Deploy a Streamlit-based interface for real-time user interaction and instant valuations.

04

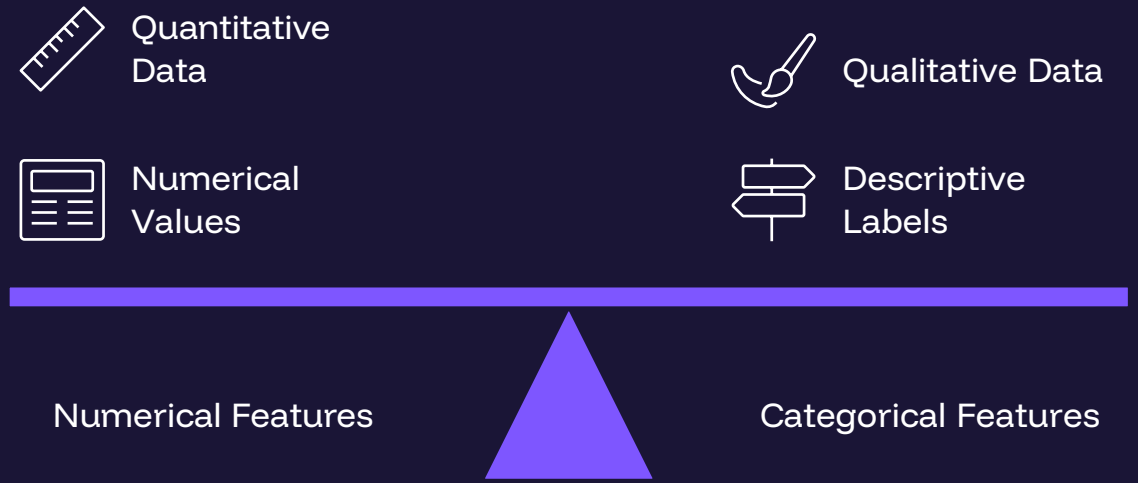
## AI Advisor

Implement an intelligent system to provide property highlights and investment ratings.

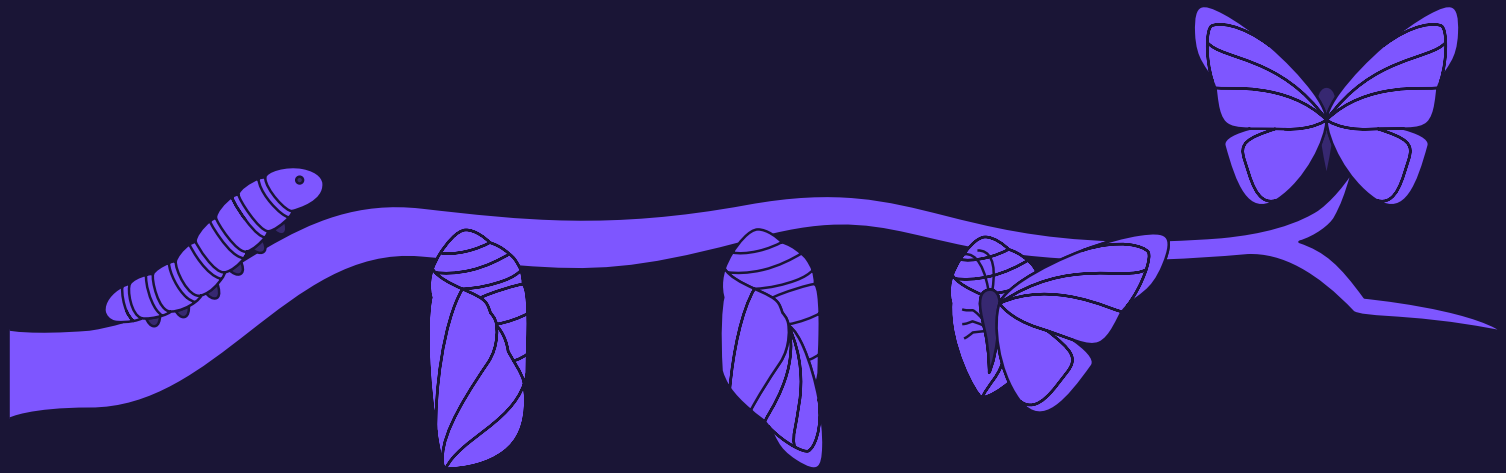
# Dataset Architecture

Comprehensive property data across major Indian cities

- Numerical: Area, Bedrooms, Distance, Crime, AQI
- Categorical: City, Locality, Property Type, Furnishing
- Binary: Lift, Security, Gym, Pool, Garden, Power
- Target: Price in Lakhs (INR)



# Property Data Analysis Pipeline



**Raw Data**

Multi-city property records

**Preprocessing**

Cleaning and pipeline construction

**EDA**

Statistical analysis and visualization

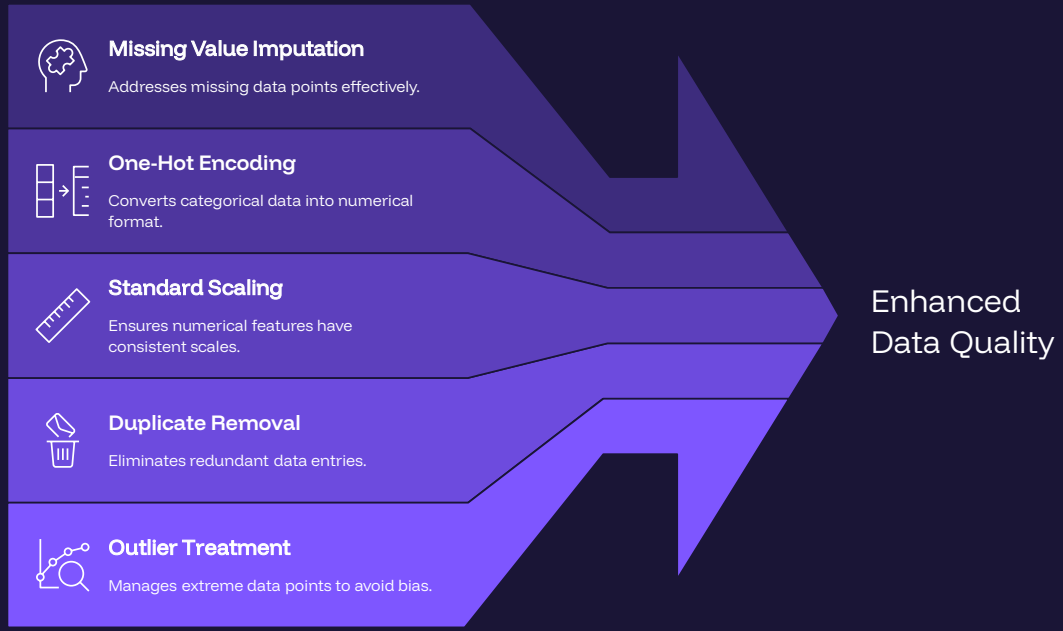
**Modeling**

Training Gradient Boosting Regressors

**Deployed App**

Streamlit Cloud application

# Data Pipeline & EDA




## EDA Insights

- Correlation between Amenities Score and Price
- Impact of City Center distance on valuation

# Algorithm Comparison

| Characteristic | Working Principle | Advantages | Selection |
| --- | --- | --- | --- |
| Linear Regression | Statistical baseline | Interpretability | Baseline |
| Random Forest | Ensemble Bagging | Handles non-linearity | Candidate |
| Gradient Boosting | Sequential Boosting | Highest Accuracy | Final Model |

# Model Performance Results

Evaluating the accuracy and robustness of our predictive valuation model.

01

## High $R^2$ Score

The model achieved a high $R^2$ score on the test dataset. This indicates strong predictive accuracy for property valuations.

02

## Diverse Property Handling

The system demonstrates robust capabilities in managing various property types. It maintains consistency across different real estate categories.

03

## Geographic Accuracy

Valuations remain precise across both urban and rural tiers. The model effectively accounts for regional market variations.

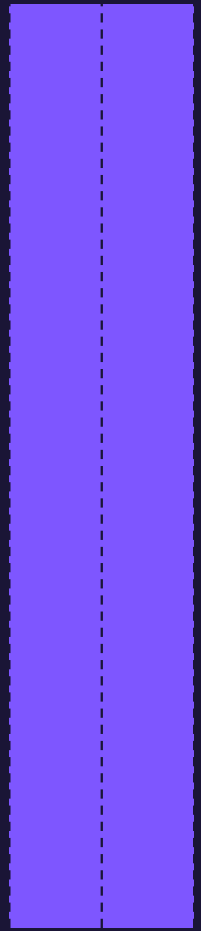
04

## Model Deployment

The final Gradient Boosting Regression model is deployed via Streamlit. This provides an accessible interface for real-time predictions.

# Live Application Dashboard

- Home Screen
- Glassmorphism UI
- Input Form
- Interactive Widgets
- Result Screen
- AI Advisor Insights



### Home Screen

The initial screen of the application

### Glassmorphism UI

The visual style of the application

### Input Form

The screen where users enter data

### Interactive Widgets

Elements that respond to user input

### Result Screen

The screen displaying the application's output

### AI Advisor Insights

AI-driven recommendations for users

# Conclusion

HomeWorth AI successfully bridges the gap between complex market data and user-friendly property valuation.

- End-to-end ML lifecycle completed
- Gradient Boosting identified as optimal regressor
- Interactive deployment for real-world utility
- Thank You! Questions?

